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Scandinavian Journal of Occupational Therapy

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/iocc20

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Vera C. Kaelin, Ingeborg Nilsson & Helena Lindgren

To cite this article: Vera C. Kaelin, Ingeborg Nilsson & Helena Lindgren (2024) Occupational therapy in the space of artificial intelligence: Ethical considerations and human-centered efforts, Scandinavian Journal of Occupational Therapy, 31:1, 2421355, DOI: 10.1080/11038128.2024.2421355

To link to this article: https://doi.org/10.1080/11038128.2024.2421355

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Published online: 08 Nov 2024.

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SCANDINAVIAN JOURNAL OF OCCUPATIONAL THERAPY
2024, VOL. 31, NO. 1, 2421355
https://doi.org/10.1080/11038128.2024.2421355

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Occupational therapy in the space of artificial intelligence: Ethical considerations and human-centered efforts

Vera C. Kaelin[a], Ingeborg Nilsson[b] and Helena Lindgren[a,b]

[a]Department of Computing Science, Umeå University; [b]Department of Community Medicine and Rehabilitation, Occupational therapy, Umeå University, Umeå, Sweden

ABSTRACT

Background: Artificial intelligence (AI) technology is constantly and rapidly evolving and has the potential to benefit occupational therapy (OT) and OT clients. However, AI developments also pose risks and challenges, for example in relation to the ethical principles of OT. One way to support future AI technology aligned with OT ethical principles may be through human-centered AI (HCAI), an emerging branch within AI research and developments with a notable overlap of OT values and beliefs.

Objective: To explore the risks and challenges of AI technology, and how the combined expertise, skills, and knowledge of OT and HCAI can contribute to harnessing its potential and shaping its future, from the perspective of OT's ethical values and beliefs.

Results: Opportunities for OT and HCAI collaboration related to future AI technology include ensuring a focus on 1) occupational performance and participation, while taking client-centeredness into account; 2) occupational justice and respect for diversity, and 3) transparency and respect for the privacy of occupational performance and participation data.

Conclusion and Significance: There is need for OTs to engage and ensure that AI is applied in a way that serves OT and OT clients in a meaningful and ethical way through the use of HCAI.

ARTICLE HISTORY
Received 1 June 2024
Revised 1 October 2024
Accepted 22 October 2024

KEYWORDS
Client-centered; healthcare; human-centred artificial intelligence; activity; participation; technology

Introduction

The use of artificial intelligence (AI) technology is constantly and rapidly evolving in industry and healthcare and has become increasingly integrated into our daily lives such as through virtual agents guiding us in phone calls or apps counting our step. AI as a concept is used for several purposes that may cause confusion. Lately, the main focus of AI has been on its technological applications, with machine learning being one of many examples. In this context, AI can be defined as 'a system's ability to correctly interpret external data, to learn from such data, and to use those learnings to achieve specific goals and tasks through flexible adaptation' [1] (p. 15). However, AI is also the research field that focuses on exploring human cognition and building theory about human capabilities through novel computational models and theories [2][1]

AI technology holds significant potential to benefit occupational therapy (OT) and OT clients. For instance, an interactive and personalised agent could support OT clients between therapy sessions by reminding them of their occupational and participation goals and goal-attainment strategies. Additionally, AI could be used to automatically gather information about a client's performed occupations from sensor data, which would then populate a weekly calendar view to be used when discussing their occupational balance with an OT [3]. While these possibilities may (or may not) sound exciting, such AI developments not only hold significant potential to benefit OT and OT clients but also pose multiple risks and challenges, also in relation to OTs' ethical principles.

This discussion paper investigates such risk and challenges of AI developments from the perspective of the World Federation of Occupational Therapists (WFOT)'s guiding principles for ethical OT [4] and discusses opportunities to engage in human-centered AI (HCAI) [5–9] as a way to shape future AI technology and benefit from its significant potential.

CONTACT Vera C. Kaelin Department of Computing Science, Umeå University, MIT-Huset, 901 87 Umeå, Sweden.

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HCAI is an emerging branch within AI development and research that challenges the historically prevalent machine-centric view of AI by placeing a strong emphasis on ensuring that AI systems are developed and deployed in a manner that benefits and respects human users (e.g. OTs and OT client), their privacy, values, and needs [5–10]. In HCAI, the focus is on designing AI to support humans in their own performance and more informed decision-making, rather than replacing human or dictating their actions [8]. Additionally, HCAI prioritises *explainable* AI systems that 'include values, ethics and privacy as core design considerations' [8] (p. 31), as opposed to, for example, unpredictable 'black box' systems that are not understood by humans [8]. In other words, Nowak et al. [8] (p. 27) define HCAI as AI that has a focus on 'collaborating with humans, enhancing human capabilities, and empowering humans to better achieve their goals'. This definition and focus of HCAI aligns remarkably well with the WFOT's guiding principles for ethical OT [4], which include 10 OT values and beliefs: i) occupation focus, ii) collaborative approach, iii) duty of care, iv) human rights and occupational justice, v) respect for diversity, vi) integrity, vii) confidentiality, viii) competency and lifelong learning, ix) obligation to the profession, and x) responsibility towards local and global societies.

To ensure that AI is developed and employed in an ethical and meaningful manner from an OT perspective, this discussion paper focuses on following questions:

1. What are the risks and challenges in AI technology from the perspective of WFOT's guiding principles for ethical OT?
2. How could OT and HCAI benefit from each other's expertise, skills, and knowledge to contribute to the development of future AI technology that aligns with OT values and beliefs per the WFOT's guiding principles for ethical OT?

To explore these questions, we reflect on the 10 OT [4] values and beliefs of the WFOT's guiding principles for ethical OT in relation to: 1) the risks and challenges posed by AI technology, 2) the potential of combining HCAI and OT expertise, skills, and knowledge to address these risks and challenges, and 3) the resulting implications for OTs' competencies, lifelong learning, as well as the profession's obligation and responsibility for the society. To structure these reflections, we have clustered the 10 OT values and beliefs into four groups, based on their connection to HCAI principles and resulting implications for OTs and the OT profession.

The first three groups, covering 7 out of the 10 OT values and bliefs [4], are discussed in the following two sections of the 'Main Text/results'. In the first section we outline risks and challenges posed by AI technology in relation to:1) occupation focus, collaborative approach, and duty of care (OT values and beliefs #i-iii), 2) occupational justice and respect for diversity (OT values and beliefs #iv-v), and 3) integrity and confidentiality (OT values and beliefs #vi-vii). In the second section, we then describe how the combined expertise, skills, and knowledge from the HCAI and OT community could address these risks and challenges and contribute to harnessing the potential of AI technology in a meaningful and ethical manner. The fourth group of OT values and beliefs of the WFOT's guiding principles of ethical OT is integrated in our 'Discussion' section to relflect on the 'Main Text/results' in regards to resulting implications for OTs and the OT profession related to OTs' competency and lifelong learning, their obligation to the profession, and responsibility towards local and global societies (OT values and beliefs #viii-x).

Main text/results

Risks and challenges in AI technology from an OT perspective

Occupation focus, collaborative approach, and duty of care

In OT, performing and participating in occupations is considered essential for individuals' health and well-being [11–14]. According to the WFOT [4], OTs safely support the occupational performance and participation of individuals, groups, and communities. This support should be provided in partnership with individuals, groups, and communities, taking into account their values, preferences and abilities.

While some developed AI systems may positively impact individuals', groups', and communities' occupational performance and participation in society (e.g. text to speech tools [15]), the disability community has raised concerns about many technological innovations and developments not meeting their needs when developed in isolation by abled bodied individuals [16,17]. Furthermore, recent reviews related to the use of AI in participation-focused paediatric rehabilitation revealed poor representation of the concept 'participation' [18,19]. Additionally, most of the found participation-focused interventions integrating AI focused on 'training' child and youth participation [20], an idea that has been challenged by the disability community and OT literature, given the importance of

the environment when supporting a client's participation in occupation [21–24].

Occupational performance can be defined as 'the client's doing' such as dressing or eating [14] and participation in occupations can be defined as 'attending' occupations and 'feeling involved' while attending these occupations [25,26]. Both, occupational performance as well as participation, are considered complex constructs, given, for example, the individual differences in how occupations are performed or how participation is experienced [14,25–27]. In addition, what is considered an occupation (i.e. meaningful activity), the strengths and challenges in occupational performance and/or participation, and the types of support an individual, group or community may require and/or prefer can vary. These individual differences require a client-centered or collaborative approach, representing another key value in OT practice and research [14,27,28]. Client-centered practice is a collaborative process between the client (individual, group, community) and the OT, where the OT recognises the client as the expert on their own situation [14]. A client-centered approach includes involving clients in decision-making processes and advocating for their needs on an individual and social level [14,29].

A lack of including people with disabilities into developments, the complexity related to occupational performance and participation, and the need for a highly personalised approach in OT can challenge current technological developments [18,19,30–32]. Even though AI is considered to deal with complex issues and has reached successful results in complex topics such as automated text generation (e.g. ChatGPT [33]), AI models require concepts to be translated in a way that computers can understand it. This can create tensions between the disability community's and OTs' desire to address complexity and existing technical requirements [18,19,31]. From an OT perspective, simplifying complex concepts is challenging due to the risk of losing important information, while technicians often shy away from the complexity of OT concepts because of the technical difficulties involved in addressing such complexity. To ensure AI developments are meaningful for the OT and OT clients, questions arise such as: How should complex OT concepts and values (i.e. occupational performance, participation, client-centredness) be translated into machine-readable information without loosing important information and while ensuring AI technology is meaningful for OT and OT clients?

Occupational justice and respect for diversity

Occupational justice 'recognizes occupational rights to inclusive participation in everyday occupations for all persons in society, regardless of age, ability, gender, social class, or other differences' [13] (p. 57). This includes the need for respecting 'the cultural diversity, lifestyles and perspectives of people' [4] (p. 4).

While AI has the potential to support tasks and accessibility such as through automated language translations [33], it can also hinder occupational performance and participation. For example, AI may replace activities that hold meaning for certain individuals and enable their participation in society (e.g. re/integration into workforce), like assembling and wrapping products, now managed by robots, or answering and transeferring phone calls, now handled by AI agents.

Furthermore, risks and challenges related to occupational justice [11–13,34] and respect for diversity arise from the predominant focus of current AI tools on majority groups, leading to bias and discrimination against marginalised groups such as people with disabilities [35]. For example, AI-powered hiring platforms have been shown to discreminate against people with facial difference due to automated detection and scoring of 'positive facial expression' which rely on algorithms trained on data lacking diversity [36]. Taken together, the questions from an OT perspective include: How can it be ensured that AI does not take over activities that are meaningful to certain individuals? How can AI-tools be made accessible and designed for all, including individuals with disabilities?

Integrity and confidentiality

Occupations are strongly connected to individuals' identities, a fact that OTs have long recognised [37]. As a result, OTs utilise this power of occupations both as a means and end in their services [12,14]. In the context of AI development, it is important to reflect on how these important and powerful data on occupational performance and participation are and should be treated in relation to OTs' obligation for integrity and guaranteeing the confidentiality of individuals' personal information [4]. Nowadays sensor-based data captured with mobile phones can be used to predict types of some activities (e.g. walking or biking) [38–41]. Occupational performance data may enable automated recognition of performed occupations, which may facilitate the automated generation of weekly plans to use in occupational therapy practice [3]. However, such data and predictions can

also be utilised by businesses to tailor advertisements towards human's activity patterns, serving marketing purposes rather than the individual or society as a whole. In addition, these data are often collected without users' active awareness on their further use (e.g. activity prediction), which can cause ethical concerns from an OT perspective and may indicate need to advocate for a better protection of clients' occupational performance and participation data. Therefore, the question arises: How can occupational performance and participation data be safeguarded and what role do OTs play in this process?

Harnessing potential and shaping future AI technology with HCAI and OT expertise

Occupation focus, collaborative approach, and duty of care

HCAI systems aim to 'focus on enhancing human capabilities allowing people to improve their own performance and successfully handle more complex tasks' [8] (p. 31). In addition, HCAI focuses on 'systems that empower humans to make more informed decisions and help them harness and channel their creativity' [8] (p. 31).

HCAI's and OT's shared focus on (occupational) performance and importance of engaging individuals, groups or communities into the development and research process (i.e. participatory research, co-designing approaches) [42–46], might provide an common ground for addressing the challenge of translating complex OT constructs and values into machine-readable information. Acknowleging the importance for co-creation and thus, valuing and integrating multiple perspectives into a research process, may facilitate the process of breaking down OT constructs and represent them in machine-readable formats. The different skills, knowledge, and expertise in OT and HCAI may also enhance each others work (see Table 1). For example, current HCAI has primarily focused on populations and legal aspects of using AI, such as laws and regulations [5–8], and biases within AI-based developments [6,8]. However, it also holds the potential to focus on the individual such as an agent collaborating with an individual to plan their upcoming week. To further develop this focus on individuals in HCAI research and development, OT theories and methods might be beneficial. Additionally, the societal-level focus in HCAI research and development could further enrich the occupational science community.

In addition, an occupational perspective may help expand the visionary statement of HCAI about 'enhancing human capabilities allowing people to improve their own performance' [8] (p. 31) to also include a focus on participation. While occupational performance and participation are highly related, they are distinct constructs. OTs are proned to help operationalise these two constructs to support their translation into machine-readable information models that can be used in AI methods, while ensuring that the needed complexity is represented and that these models are used in a meaningful manner.

Occupational justice and respect for diversity

According to Nowak and colleagues [8], 'the concept of HCAI envisions future AI technology that will synergistically work with humans for the benefit of humans and human society' (p. 31), including 'neutralizing the bias introduced by the algorithm' (p. 32). In other words, HCAI is not primarily driven by financial gain but rather by serving humans and society as a whole [5–8]. This includes developing and implementing AI systems in a manner so that they serve all individuals, groups and communities, including minority groups.

Complementing these HCAI efforts with an OT perspective on the link between occupations and an individuals' health and well-being, may foster important reflections and discussions on the consequences when AI replaces occupations or hinders re/integration into workforce. This may result in the need to carefully evaluate and advocate for individuals' preferences about when and in what way AI should support them.

Furthermore, collaborative efforts between HCAI and OT can contribute to ensuring that AI technology is not only developed for the majority group but also for minority groups such as people with disabilities, whose data are often underrepresented in algorithms, leading to unequal opportunities and discrimination. The disability community has emphasised the necessity for not only human-centered AI but also disability-centered AI [47]. Disability-centered AI focuses on designing AI systems specifically with and for people with disabilities [47]. This approach underscores the importance of involving people with disabilities in AI system development, as their reality differs from non-disabled humans. People with disabilities often encounter environments that are not designed for their needs, such as the prior mentioned AI-powered hiring platforms that are not designed to recognise facial experessions of individuals with facial differences [36]. To ensure AI systems are designed with and for people with

Table 1. HCAI And OT contributions and potential benefits for collaboration.

OT Values and Beliefs	HCAI Contribution	OT Contribution	HCAI and OT Collaboration
Occupation focus, collaborative approach, and duty of care	HCAI's focus: • Enhancing human capabilities through HCAI, allowing people to improve their own performance and successfully handle more complex tasks [6,8,9]. • AI systems that empower humans to make more informed decisions and help them harness and channel their creativity [6,8,9]. Skills HCAI experts bring: • Design methodologies targeting values in HCAI [6,8,9,52,54]. • Expertise on using AI on a population level, including related legal aspects of using AI (e.g. laws and regulations) and introduced biases within AI-based developments [5–8]. • Technical knowledge and perspective on enhancing human capabilities [5–8].	OT's focus: • Occupational performance and participation, and its relation to health and well-being [11–13]. • The need for and implementation of client-centeredness [14,27,28]. Skills OT experts bring: • Co-creating research and knowledge about OT-relevant topics with OT clients [42–45]. • In-depth knowledge about client-centeredness and occupational performance and participation on an individual and societal level, including factors influencing it, [11–14,27,28]. • Fundamental research methodologies necessary for HCAI.	• Develop tools, together with OT clients, focusing on occupational performance and participation, and, thus, ensure meaningfulness for OT clients. • Together, tackle occupation and participation needs with a focus on the individual (e.g. OT clients) as well as societal level (e.g. legal aspects) and responsible, human-centered AI. • Include a specific focus on participation in HCAI's mission for future work.
Occupational justice and respect for diversity	HCAI's focus: • AI systems for the benefit of humans and human society [5–8]. • Fair AI systems that reduce bias [6,8,52]. Skills HCAI experts bring: • Skills related to biases in data, and their consequences for minority groups [6,8,35,52,55].	OT's focus: • Ethics related to occupational justice [4,11–13,34]. • Ethics related to data from vulnerable populations, consequences of biases [4]. Skills OT experts bring: • An occupational justice lens including knowledge related to occupation, health and justice [11–13,34] • Advocacy and collaboration with the disability community [42–45,56].	• OTs know about occupational performance and participation and ethics related to data and the use of AI-enabled system such as assistive devices using text to speech features. HCAI scientists and developers know about privacy and security, implementing fair accessibility of AI functionalities.
Integrity and confidentiality	HCAI's focus: • Explainable, transparent, validated, and thus trustworthy systems [8,52]. • Values, ethics, and privacy as core design considerations in AI systems and applications [8,52]. Skills HCAI experts bring: • Technical skills such as coding. • Skills related to how to protect data from a technical point.	OT's focus: • Ethics related to occupational performance and participation data. Skills OT experts bring: • Know about occupational performance and participation data and the power that comes with it [14,37]. • Knowledge about how to treat sensitive data [4]. • Diversity and different abilities people have and needs they may have to understand how data will be used.	• Foster discussions regarding the benefits and risks of collecting OT-relevant data in electronic forms, and how related pros and cons should be communicated to clients. • Provide information in an accessible way (e.g. through different literacy levels and accessible technology) to support informed decision-making among OT clients. • Knowledge about the sensitive character of data and how to store them in an electronically safe and protected way, to ensure individuals' privacy.

disabilities, OTs and HCAI researchers may be important stakeholders to collaborate with the disability community to advocate for disability-centered AI [47] alongside human-centered AI.

Integrity and confidentiality

HCAI aims to 'focus on explainable, transparent, validated, and thus trustworthy systems [that] include values, ethics, and privacy as core design considerations [to optimally support] both individuals and society as a whole in dealing with the increasing complexity of a networked, globalised world'. [8] (p. 31). These efforts are aligned with ongoing discussions in medical ethics, which stress the importance of explainable, transparent, valid and trusthworthy AI in healthcare systems [48,49]. This includes the need for AI applications to be understood by humans, for users to know and control how their data are used, and ensuring that data are not transferred to any third-parties without user consent. For example, an OT should be able to comprehend an AI system's decision-making process and why it researched certain conclusions, requiring alternatives to black box models. It also includes ensuring that systems handling sensitive data, such as individual's occupational performance and participation data, refrain from communicating with each other unless the individuals actively instructs the system to do so.

Complementing HCAI efforts by incorporating an OT perspective could foster important discussions regarding the benefits and risks of collecting OT-relevant data in an electronic form, as well as how these pros and cons should be communicated to clients. It is important to ensure that users such as OT clients understand these instructions as well as the AI tools itself to make sure they can make an informed decision. This includes providing instructions in inclusive ways, such as accommodating for different abilities and needs among people in our society (e.g. different literacy levels). While OTs and occupational scientists, together with the disability community could support with knowledge on accommodations for diversity in our society, HCAI scientists and developers could support with technological advancements for making this possible.

Discussion

The use of technology, including AI, is considered a research priority by organisations such as WFOT [50] and the National Institutes of Health [51]. From an OT perspective, relevant questions related to this research priority include 'what kind of AI do we as OTs want to bring forward?', 'should OTs be involved in HCAI developments and research or hand it over to designers and doctors?', and 'how can be ensured that AI technology aligns with principles of ethical OT?' To our knowledge, this is one of the first papers discussing the engagement of OT in responsible and trustworthy AI developments and implementation to support OT and the occupational performance and participation of OT clients. We do so by reflecting on risk and challenges of AI technology from an OT ethics perspective and how OT and HCAI could complement each other and address such risks and challenges. For the discussion of this paper, we will focus on implications for OTs and the OT profession related to the following values and beliefs of the WFOT's guiding principles for ethical OT: viii) OTs' competency and lifelong learning, ix) obligation to the profession, and x) responsibility towards local and global societies.

We believe that the different expertise in OT and HCAI (e.g. knowledge and skills related to OT concepts versus technological aspects) and the remarkable overlap between the underlying principles of OT and HCAI (e.g. client- and human-centeredness [8,27,28]), makes them a promising combination. Together, OTs and HCAI developers and researchers can ensure that AI systems are developed and implemented in a responsible manner such as through collaboration with end users (e.g. OT clients and/or practitioners – depending on the focus of the AI system) and by adhering to ethical AI guidelines (e.g. explainability, fairness, transparency, validation and, thus, trustworthiness of AI systems) [52].

To support HCAI efforts from an OT perspective and help bringing forward the type of AI that aligns with OTs' ethical values and beliefs [4], we belief that it is OTs' obligation to the profession and towards the local and global societies to start to actively engage in discussions within and outside the OT community on how and in what situations (e.g. for what challenges) AI could be a supportive technology that serves OT and OT clients in a meaningful and ethically responsible way. This includes integrating discussions related to OT and HCAI into the OT curriculum and into further education for OT practitioners. Discussion and knowledge related to HCAI can heighten awareness within the OT community about the risks and benefits of AI development, use, and implementation. It facilitates the recognition of developments heading in the wrong direction, such as lacking a trustworthy approach or failing to integrate the end-user community, which may prevent a user's developing distrust,

fear and avoidance, as well as the risk of creating a tool that is neither used nor needed.

One way to support HCAI efforts with an OT perspective is through involvement in HCAI research at universities' research labs when OT-relevant tools are being developed and tested such as automated detection of human occupations and digital agents supporting individuals occupational performance and participation. However, technological developments do not only happen in research labs, but often also in practice such as when clinics transition to an electronic data capturing system involving AI. We believe that such situations are important opportunities for OTs to get involved and ensure that anticipated AI developments are meaningful for the profession and OT clients. This includes that AI functionalities are tailored to specific needs, which may differ between OTs (e.g. how does data need to be captured that therapists can work with it) and OT clients (e.g. whether the AI system are inclusive and accessible for minority groups such as people with disabilities). Additionally, OT engagement in such transitions might be relevant to ensure that data on occupational performance and participation are included in these data capturing systems in a meaningful and responsible way. This may require active and critical questioning of the (to be) developed and implemented system to find out about its trustworthiness including whether ethical aspects are considered and respected. It may also require advocacy to ensure that OT clients are clearly informed about the data use and storage in these systems. To support such reflection and active questioning of the trustworthiness of AI systems, the European Union has developed the Assessment List for Trustworthy AI (ALTAI) [53], a checklist to help raise awareness regaring the potential societal, environmental, workforce, and citizen impacts of AI. Question and reflection in this paper may provide an OT perspective to such an assessment. Similarly, they may contribute, together with the ALTAI tool, to include a technology focus into future guidelines for ethical OT and may lay the basis for a research agenda on the use of responsible and trustworthy AI in OT.

Conclusion

Regardless of whether AI-based developments excite or scare us as OTs, they are progressing at an increased speed and influence our profession and services for our clients. We strongly believe that these developments will not just stop. Thus, we as OTs should engage in these developments and ensure that AI is applied in an responsible way that aligns with our ethical guidelines and our values, so that it serves us and our clients in a meaningful and ethical manner through the use of HCAI. This includes the need for OTs to actively engage in AI development research initiatives as well as advancements in practice such as the transition to electronic data capturing systems, which often provide data points for AI systems. Furthermore, it underscores the importance of integrating HCAI education into the OT curricula to ensure that future OTs are equipped with the knowledge and skills necessary to engage in AI system development in research and practice.

Note

1. For more meanings, see van Rooij, et al. (2024) Reclaiming AI as theoretical tool for cognitive science. https://osf.io/preprints/psyarxiv/4cbuv).

Acknowledgements

The content of this manuscript does not necessarily represent the policy of NIDILRR, ACL, or HHS, and you should not assume endorsement by the Federal Government. We thank Dr. Mary Khetani for her support during the initial brainstorming that contributed to the development of this article.

Disclosure statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Funding

This work was supported with funds from the National Institute on Disability, Independent Living, and Rehabilitation Research (90SFGE0032-01-00, PI: V. Kaelin). NIDILRR is a Center within the Administration for Community Living (ACL), Department of Health and Human Services (HHS).

ORCID

Vera C. Kaelin http://orcid.org/0000-0003-1290-9441
Ingeborg Nilsson http://orcid.org/0000-0002-8265-5769
Helena Lindgren http://orcid.org/0000-0002-8430-4241

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